

COMPUTER NETWORKS

Unit-III: OSI Reference Model and Network Connectivity

Comprehensive Study Notes

3.1 Introduction to OSI Reference Model

Theory

The **OSI (Open Systems Interconnection) Reference Model** is a conceptual framework developed by the **International Organization for Standardization (ISO)** in 1984. It standardizes how different computer systems communicate over a network.

Why OSI Model?

The OSI model was developed to address the need for a standardized networking framework that ensures interoperability between different vendors' equipment. It divides network communication into seven distinct layers, each with specific functions.

Layers Overview

The OSI model has seven layers: Physical (Layer 1), Data Link (Layer 2), Network (Layer 3), Transport (Layer 4), Session (Layer 5), Presentation (Layer 6), and Application (Layer 7).

Mnemonic to Remember OSI Layers

Mnemonic	Layer
All	Application
People	Presentation
Seem	Session
To	Transport

Mnemonic	Layer
Need	Network
Data	Data Link
Processing	Physical

3.2 OSI Physical Layer (Layer 1)

Theory

The **Physical Layer** is the lowest layer of the OSI model. It deals with the **physical transmission** of raw bit streams over a physical medium.

Functions of Physical Layer

The Physical Layer is responsible for bit synchronization, bit rate control, physical topologies, transmission mode determination, and the physical representation of data (encoding). It defines the electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating the physical link.

Key Concepts

Concept	Description	Example
Data Encoding	Representing bits as signals	Manchester encoding
Bit Rate	Speed of data transfer	100 Mbps, 1 Gbps
Transmission Media	Physical path	Copper, Fiber, Wireless
Connector Types	Physical interfaces	RJ45, SC, ST

Transmission Modes

Mode	Direction	Example
Simplex	One way only	TV broadcast
Half-Duplex	Both ways, one at a time	Walkie-talkie

Mode	Direction	Example
Full-Duplex	Both ways simultaneously	Telephone

3.3 Data Link Layer (Layer 2)

Theory

The **Data Link Layer** is responsible for **node-to-node** delivery of data. It ensures reliable transmission of frames between directly connected nodes.

Functions of Data Link Layer

Key functions include framing (dividing data into frames), physical addressing (MAC addresses), error control (detecting and retransmitting damaged frames), flow control (regulating data flow), and access control (determining which device can use the channel at a given time).

Sub-Layers

The Data Link Layer is divided into two sub-layers:

Sub-Layer	Function
LLC (802.2)	Error and flow control, addressing
MAC (802.3)	Media access, physical addressing

Frame Structure

```
+-----+
+
| Header | Data | Trailer |
+-----+-----+-----+
| Dest MAC | Src MAC | Data | FCS |
| (6 bytes) | (6 bytes) | (46-1500) | (4 bytes) |
+-----+-----+-----+-----+
```

3.4 Network Layer (Layer 3)

Theory

The **Network Layer** is responsible for **host-to-host** delivery of packets across multiple networks. It handles **routing** and **logical addressing**.

Functions of Network Layer

The Network Layer handles logical addressing (assigning IP addresses), routing (determining the best path for data), packet forwarding, fragmentation and reassembly, and network congestion control.

Key Concepts

Concept	Description	Example
Logical Addressing	Unique address for each host	IP Address: 192.168.1.1
Routing	Path selection	RIP, OSPF, BGP
Packet Forwarding	Sending packets	Router operation

Routing Types

Routing can be classified as static (manual configuration) or dynamic (automatic route discovery using protocols such as RIP, OSPF, BGP). Dynamic routing can further be distance vector, link state, or path vector.

3.5 Transport Layer (Layer 4)

Theory

The **Transport Layer** is responsible for **end-to-end** communication and ensures complete delivery of data.

Functions of Transport Layer

The Transport Layer provides service-point addressing (ports), segmentation and reassembly, connection control (connection-oriented or connectionless), flow control, error control, and congestion control.

Transport Protocols

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Flow Control	Yes	No
Error Control	Yes	No
Speed	Slower	Faster

3.6 Session Layer (Layer 5)

Theory

The **Session Layer** establishes, manages, and terminates **sessions** (connections) between applications.

Functions

The Session Layer handles session establishment, maintenance, and termination; dialog control (half-duplex or full-duplex); synchronization (inserting checkpoints for recovery); and token management (controlling which side can transmit).

Key Features

Feature	Description
Session Establishment	Setting up communication
Dialog Control	Half/Full duplex control
Synchronization	Checkpoints for recovery
Token Management	Who can transmit

3.7 Presentation Layer (Layer 6)

Theory

The **Presentation Layer** ensures that data is in a **readable format** for the application layer. It handles **data formatting**, **encryption**, and **compression**.

Functions

This layer is responsible for translation (converting between different data formats), encryption/decryption for security, and compression to reduce data size before transmission.

Data Formats

Format Type	Example
Text	ASCII, Unicode, EBCDIC
Image	JPEG, GIF, PNG
Audio	MP3, WAV
Video	MPEG, AVI

3.8 Application Layer (Layer 7)

Theory

The **Application Layer** is the highest layer. It provides services directly to **end-user applications**.

Functions

The Application Layer provides network services to user applications, including file transfer, email, web browsing, remote login, and directory services. It identifies communication partners, determines resource availability, and synchronizes communication.

Common Application Layer Protocols

Protocol	Port	Purpose
HTTP	80	Web browsing
HTTPS	443	Secure web
FTP	20, 21	File transfer
SMTP	25	Email sending
POP3	110	Email receiving
DNS	53	Name resolution
TELNET	23	Remote access

3.9 TCP/IP Model

Theory

The **TCP/IP Model** is the practical implementation of the OSI model, developed by the US Department of Defense.

TCP/IP Architecture

The TCP/IP model consists of four layers: Application Layer (combines OSI layers 5-7), Transport Layer (layer 4), Internet Layer (layer 3), and Network Access Layer (combines layers 1-2).

TCP/IP vs OSI

Feature	OSI Model	TCP/IP Model
Layers	7 Layers	4 Layers
Development	ISO (1984)	DOD (1970s)
Protocols	Theoretical	Practical
Implementation	Reference model	Real-world use

TCP/IP Protocol Suite

The TCP/IP suite includes protocols at each layer: Network Access (Ethernet, WiFi), Internet (IP, ICMP, ARP), Transport (TCP, UDP), and Application (HTTP, FTP, SMTP, DNS).

3.10 Network Devices

Theory

Network devices are hardware components that connect and manage network traffic.

3.10.1 Repeater

Theory: A repeater regenerates and retransmits signals to extend the network distance.

Features: Works at **Physical Layer** (Layer 1); **Amplifies** and **regenerates** signals; extends network distance; **does not filter** data.

Advantages: Extends network range; simple and inexpensive; **no processing delay**.

Disadvantages: Amplifies **noise** along with data; **does not connect** different network types; limited to single network.

3.10.2 Hub

Theory: A hub is a central connecting device that broadcasts data to all connected devices.

Type	Features	Use
Passive Hub	Just connects wires	Small networks
Active Hub	Amplifies signals	Larger networks
Intelligent Hub	Management features	Enterprise networks

Disadvantages: **Collision domain** — all devices share; **Bandwidth sharing**; **Security issues** (broadcasts); **Obsolete** (replaced by switches).

3.10.3 Bridge

Theory: A bridge connects two network segments and filters traffic based on MAC addresses.

Type	Description	Use
Local Bridge	Connects local segments	LANs in same building
Remote Bridge	Connects distant segments	LANs in different cities
Wireless Bridge	Wireless connection	Hard-to-wire areas

Features: Works at **Data Link Layer** (Layer 2); **Intelligent** — decides which frames to forward; reduces **collision domains**; **Transparent** to network layer.

3.10.4 Switch

Theory: A switch is a multi-port bridge that improves network performance by directing data only to the destination.

Types of Switches Based on Function

Type	Function	Use
Unmanaged	Simple switching	Home/Small office
Managed	Configuration	Enterprise
Smart	Basic management	SMB
PoE	Power over Ethernet	Cameras, Phones
Layer 3	Routing capability	Large networks

Switch Operation

MAC Address Table:

+-----+		
	MAC Address	Port
+-----+		
	AA:AA:AA:AA:AA:AA	1
	BB:BB:BB:BB:BB:BB	2
	CC:CC:CC:CC:CC:CC	3
+-----+		

Features: **Each port** is a collision domain; **Full-duplex** communication; **Higher bandwidth**; **Better performance**.

Advantages over Hub: No collisions; faster speed; more secure; scalable.

3.10.5 Router

Theory: A router is a network device that forwards data packets between different networks.

Feature	Static Router	Dynamic Router
Configuration	Manual	Automatic
Routing Protocol	None	RIP, OSPF, BGP
Adaptability	No	Yes
Best for	Small networks	Large networks

Routing Protocols

Protocol	Type	Use
RIP	Distance Vector	Small networks
OSPF	Link State	Medium networks
BGP	Path Vector	Internet routing

Router Functions

1. **Path Determination** — Finding best route
2. **Packet Forwarding** — Sending packets
3. **Network Segmentation** — Separating networks
4. **Address Translation** — NAT
5. **Security** — Firewall capabilities

3.10.6 Brouter

Theory: A brouter combines the functions of a **bridge** and a **router**.

Features: Acts as **bridge** for local traffic; Acts as **router** for remote traffic; **Flexible** and **cost-effective**.

Advantages: Reduces network traffic; cost-effective (two devices in one); **reduces complexity**.

Disadvantages: Slower than dedicated devices; complex to configure.

3.10.7 Gateway

Theory: A gateway connects networks with **different protocols** and translates between them.

Features: Works at **higher layers** (4-7); **Protocol translation**; Connects **different networks**; **Application level** processing.

Types of Gateways

Type	Function
Protocol Gateway	Translates protocols
Application Gateway	Proxy services
Voice Gateway	Voice/Data conversion

Device Comparison

Device	Layer	Function	Smart
Repeater	Layer 1	Signal amplification	No
Hub	Layer 1	Broadcasting	No
Bridge	Layer 2	MAC filtering	Yes
Switch	Layer 2	Intelligent switching	Yes
Router	Layer 3	Network routing	Yes
Brouter	Layer 2/3	Bridge + Router	Yes
Gateway	Layer 4-7	Protocol translation	Yes

3.11 Real World Networks

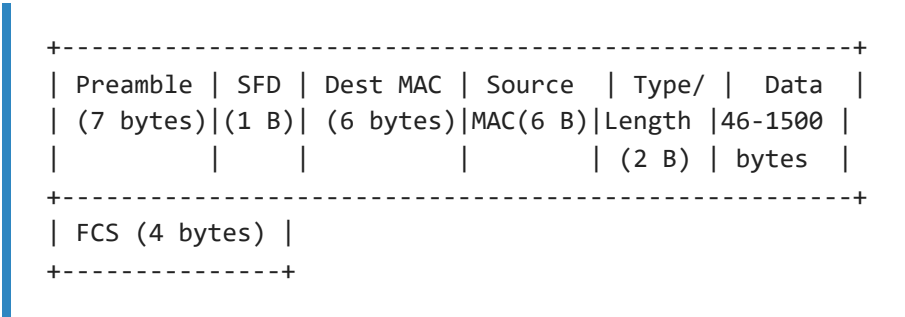
3.11.1 Ethernet

Theory: Ethernet is the most widely used LAN technology, standardized by IEEE 802.3.

Ethernet Evolution

Ethernet has evolved from 10 Mbps (10Base-T) to 100 Mbps (Fast Ethernet), 1 Gbps (Gigabit Ethernet), 10 Gbps, 40 Gbps, and 100 Gbps standards.

Ethernet Frame Format



Field	Size	Description
Preamble	7 bytes	Synchronization
SFD	1 byte	Start of frame
Dest MAC	6 bytes	Destination address
Src MAC	6 bytes	Source address
Type/Length	2 bytes	Protocol or length
Data	46-1500 bytes	Payload
FCS	4 bytes	Error detection

Ethernet Operation (CSMA/CD)

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the access method used by Ethernet. Before transmitting, a device listens (carrier sense). If the channel is idle, it transmits; if busy, it waits. If two devices transmit simultaneously, a collision occurs, and each device waits a random backoff period before retransmitting.

Ethernet Cabling & Components

Ethernet uses various cabling: twisted pair (UTP/STP with RJ45 connectors), coaxial, and fiber optic. Components include NICs, switches, routers, and cables.

3.11.2 Token Ring

Theory: Token Ring is a network technology where a special token (frame) circulates around the network. It follows the **IEEE 802.5** standard.

Features: **Token passing** access method; **Predictable access**; **No collisions**.

Advantages: **Deterministic** access; **Equal access** for all devices; **No collisions**; **Better performance** under heavy load.

Disadvantages: **Expensive**; **Single point of failure**; **Complex to manage**; **Obsolete**.

3.11.3 Token Bus

Theory: Token Bus combines the features of **token passing** and **bus topology** (IEEE 802.4). It uses a **physical bus** but forms a **logical ring** for token passing.

Characteristics: **Collision-free**; **Priority** support; **Deterministic**. Used in **manufacturing** environments.

Disadvantages: **Complex**; **Expensive**; **Rarely used**.

Token Ring vs Token Bus

Feature	Token Ring	Token Bus
Topology	Physical Ring	Physical Bus
Token Passing	Physical order	Logical order
Standard	IEEE 802.5	IEEE 802.4
Use	Office networks	Industrial

3.11.4 Network Connectivity Components

NIC (Network Interface Card)

Connects device to network; has **MAC address**; handles physical connection; provides **unique identifier**.

Physical Address (MAC Address)

48 bits (6 bytes); written in **hexadecimal**; **uniquely identifies** device.

```
MAC Address Example: AA:BB:CC:11:22:33
OUI                   | NIC specific
(First 3 bytes)       | (Last 3 bytes)
```

Logical Address (IP Address)

32 bits (IPv4) or 128 bits (IPv6); written in **decimal** format; **hierarchical** addressing; identifies network and host.

```
IPv4 Address Example: 192.168.1.100
Network   | Host
192.168.1 | 100
```

3.12 Complete Summary

OSI Layers Summary

Layer	Name	Function	Data Unit
7	Application	User interface	Data
6	Presentation	Data formatting	Data
5	Session	Connection management	Data
4	Transport	End-to-end delivery	Segments
3	Network	Routing/addressing	Packets
2	Data Link	Node-to-node delivery	Frames
1	Physical	Signal transmission	Bits

TCP/IP vs OSI

OSI Layer	TCP/IP Layer	Key Protocols
7,6,5	Application	HTTP, FTP, SMTP, DNS
4	Transport	TCP, UDP
3	Internet	IP, ICMP, ARP
2,1	Network Access	Ethernet, WiFi

Network Devices Summary

Device	Layer	Function
Repeater	Physical	Regenerates signal
Hub	Physical	Broadcasts data
Bridge	Data Link	Connects segments
Switch	Data Link	Intelligent switching

Device	Layer	Function
Router	Network	Routes between networks
Brouter	2/3	Bridge + Router
Gateway	4-7	Protocol translation

Ethernet Standards

Name	Speed	Medium	Distance
10Base-T	10 Mbps	UTP	100m
100Base-TX	100 Mbps	CAT5	100m
1000Base-T	1 Gbps	CAT5e/6	100m
10GBase-T	10 Gbps	CAT6/7	55-100m

3.13 Previous Year Questions

Short Questions (2 marks)

1. What is the OSI model?
2. Name the layers of the OSI model.
3. What is the difference between a hub and a switch?
4. What is a router?
5. Define MAC address.
6. What is CSMA/CD?
7. What is the difference between TCP and UDP?
8. What is a gateway?

Medium Questions (5 marks)

1. Explain the functions of different layers of the OSI model.
2. Compare OSI and TCP/IP models.
3. Explain Ethernet frame format.
4. What is the difference between a bridge, switch, and router?
5. Explain token ring technology.

Long Questions (12 marks)

1. (a) Draw and explain the OSI reference model with all layers.
2. (a) Explain the working of Ethernet with CSMA/CD.
3. (a) Explain different types of network devices.

Quick Reference Card

OSI Model Memory Aid

All People Seem To Need Data Processing

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

Device Functions Quick Summary

Repeater: Amplifies signals
Hub: Broadcasts to all ports
Bridge: Connects segments (MAC based)
Switch: Intelligent switching
Router: Routes between networks
Gateway: Protocol translation

Ethernet Frame Fields

Preamble | SFD | Dest MAC | Src MAC | Type/Length
| Data | FCS
7 bytes | 1 B | 6 bytes | 6 bytes | 2 bytes
|46-1500| 4 B

Key Protocols by Port

HTTP -> 80 HTTPS -> 443
FTP -> 21 SMTP -> 25

DNS	->	53	POP3	->	110
TELNET	->	23	SSH	->	22